

Variance and covariance

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Deviation, error, and noise

Definition 0.1 (Deviation). A **deviation** is the difference between a value and a reference value. For any quantity z and reference value r , the deviation of z from r is:

$$z - r$$

In probability and statistics, “deviation” usually means deviation from a [population mean](#).

See: [Wikipedia: Deviation \(statistics\)](#)

Example 0.1 (Deviation from a reference weight). A newborn weighing 3,200 g deviates from a reference weight of 3,500 g by $3200 - 3500 = -300$ g.

Definition 0.2 (Deviation from a population or subpopulation mean). The **deviation from the mean** of a random variable Y is its [deviation](#) from its [expectation](#):

$$e(Y) \stackrel{\text{def}}{=} Y - E[Y]$$

For a realized observation y :

$$e(y) \stackrel{\text{def}}{=} y - E[Y]$$

Other sources often call this quantity an **error** or **noise term**. In regression settings, the reference mean is often conditional on covariates: $e(y_i) \stackrel{\text{def}}{=} y_i - E[Y_i \mid X_i = x_i]$.

These notes prefer “deviation” for this mean-deviation quantity; “error” and “noise” are common aliases. The Morrison Lab’s regression notes use “residual” (defined in their [Linear regression chapter](#)) for deviations from fitted values, write $e(\cdot)$ for these model/data deviations, and reserve $\varepsilon(\cdot)$ for estimator-to-estimand deviations (see [Estimation](#)).

See:

- [Wikipedia: Errors and residuals](#)
- [Wikipedia: Deviation \(statistics\)](#)
- [Wikipedia: Linear regression — Notation and terminology](#)

Example 0.2 (Deviation of a die roll from its mean). A fair die roll Y has $E[Y] = 3.5$ (computed on the [expectation page](#)), so a roll of $y = 5$ has deviation $e(5) = 5 - 3.5 = 1.5$, and a roll of $y = 2$ has deviation $e(2) = 2 - 3.5 = -1.5$.

Variance and related characteristics

Definition 0.3 (Variance). The **variance** of a random variable X with $E[X^2] < \infty$ is the [expectation](#) of the squared [deviation from the mean](#); that is:

$$\text{Var}(X) \stackrel{\text{def}}{=} E[[e(X)]^2]$$

Theorem 0.1 (Variance as expected squared deviation from the mean).

$$\text{Var}(X) = E[(X - E[X])^2]$$

i Proof

Proof. Substituting the definition of $e(X)$ from Definition [0.2](#) into Definition [0.3](#):

$$\text{Var}(X) \stackrel{\text{def}}{=} E[[e(X)]^2] = E[(X - E[X])^2].$$

□

This is the more commonly seen form of the variance definition, and the canonical starting point in most treatments; it's given here as a theorem rather than the primary definition to keep $e(X)$ as the definition's single building block, matching the notation of Definition [0.2](#).

Theorem 0.2 (Simplified expression for variance).

$$\text{Var}(X) = E[X^2] - (E[X])^2$$

i Proof

Proof. By [linearity of expectation](#), with $E[X]$ treated as a constant:

$$\begin{aligned}
\text{Var}(X) &\stackrel{\text{def}}{=} \text{E}[(e(X))^2] && \text{(definition of variance)} \\
&= \text{E}[(X - \text{E}[X])^2] && \text{(definition of deviation from mean)} \\
&= \text{E}[X^2 - 2X \text{E}[X] + (\text{E}[X])^2] && \text{(expand binomial square)} \\
&= \text{E}[X^2] - 2 \text{E}[X] \text{E}[X] + (\text{E}[X])^2 && \text{(linearity of expectation; E[X] is a constant)} \\
&= \text{E}[X^2] - (\text{E}[X])^2 && \text{(combine like terms)}
\end{aligned}$$

□

Example 0.3 (Variance of a Bernoulli random variable). Let $X \sim \text{Ber}(\pi)$. Then $\text{E}[X] = \pi$ and $\text{E}[X^2] = \pi$ (both computed on the [expectation page](#)), so:

$$\begin{aligned}
\text{Var}(X) &= \text{E}[X^2] - (\text{E}[X])^2 && \text{(simplified expression for variance)} \\
&= \pi - \pi^2 && \text{(substitute)} \\
&= \pi(1 - \pi) && \text{(factor)}
\end{aligned}$$

For a fair coin ($\pi = 1/2$), $\text{Var}(X) = 1/4$.

Definition 0.4 (Conditional variance). The **conditional variance** of Y given $X = x$ is the variance of Y under its conditional distribution given $X = x$:

$$\text{Var}(Y \mid X = x) \stackrel{\text{def}}{=} \text{E}[(Y - \text{E}[Y \mid X = x])^2 \mid X = x]$$

Evaluating this function of x at X gives the random variable $\text{Var}(Y \mid X)$, just as the [conditional expectation function](#) $\text{E}[Y \mid X]$ comes from $\text{E}[Y \mid X = x]$.

Example 0.4 (Conditional variance of a binary outcome). Let (X, Y) have the joint PMF $\text{P}(X = 0, Y = 0) = 0.2$, $\text{P}(X = 0, Y = 1) = 0.3$, $\text{P}(X = 1, Y = 0) = 0.1$, $\text{P}(X = 1, Y = 1) = 0.4$ (the table in the [expectation page's exercise](#)). Given $X = 0$, Y is binary with $\text{P}(Y = 1 \mid X = 0) = 0.3/0.5 = 0.6$, so $\text{E}[Y \mid X = 0] = 0.6$, and by Definition 0.4:

$$\begin{aligned}
\text{Var}(Y \mid X = 0) &= (0 - 0.6)^2 \cdot 0.4 + (1 - 0.6)^2 \cdot 0.6 && \text{(definition of conditional variance)} \\
&= 0.144 + 0.096 && \text{(multiply)} \\
&= 0.24 && \text{(add)}
\end{aligned}$$

The same steps with $\text{P}(Y = 1 \mid X = 1) = 0.4/0.5 = 0.8$ give $\text{Var}(Y \mid X = 1) = (0.2)^2 \cdot 0.8 + (0.8)^2 \cdot 0.2 = 0.16$.

Definition 0.5 (Homoskedasticity). A random variable Y is **homoskedastic** with respect to a random variable (or vector) X if the **conditional variance** of Y given $X = x$ is the same for every x :

$$\text{Var}(Y \mid X = x) = \sigma^2 \quad \text{for all } x$$

for some constant σ^2 .

Example 0.5 (A homoskedastic total). Flip two fair coins, let X_1 indicate heads on the first, and let Y be the total number of heads. Given $X_1 = 0$, Y is 0 or 1 with probability 1/2 each; given $X_1 = 1$, Y is 1 or 2 with probability 1/2 each. Either way, Y is 1/2 away from its conditional mean with probability 1, so $\text{Var}(Y \mid X_1 = 0) = \text{Var}(Y \mid X_1 = 1) = 1/4$, and Y is homoskedastic with respect to X_1 , with $\sigma^2 = 1/4$.

Definition 0.6 (Heteroskedasticity). A random variable Y is **heteroskedastic** with respect to X if it is not **homoskedastic** with respect to X : that is, if $\text{Var}(Y \mid X = x)$ differs between some values of x .

Example 0.6 (A heteroskedastic binary outcome). In Example 0.4, $\text{Var}(Y \mid X = 0) = 0.24 \neq 0.16 = \text{Var}(Y \mid X = 1)$, so Y is heteroskedastic with respect to X . More generally, a binary outcome with $P(Y = 1 \mid X = x) = \pi(x)$ has conditional variance $\pi(x)(1 - \pi(x))$, so it is homoskedastic only if $\pi(x)$ takes at most two values, π and $1 - \pi$, for some π .

Theorem 0.3 (Law of total variance). For random variables X and Y with $E[Y^2] < \infty$:

$$\text{Var}(Y) = E[\text{Var}(Y \mid X)] + \text{Var}(E[Y \mid X])$$

i Proof

Proof. Write $\mu \stackrel{\text{def}}{=} E[Y]$ and $m(X) \stackrel{\text{def}}{=} E[Y \mid X]$. Adding and subtracting $m(X)$ inside the deviation and expanding the square:

$$\begin{aligned} \text{Var}(Y) &= E[(Y - \mu)^2] && \text{(definition of variance)} \\ &= E[(Y - m(X)) + (m(X) - \mu)^2] && \text{(add and subtract } m(X)) \\ &= E[(Y - m(X))^2] + E[(m(X) - \mu)^2] + 2E[(Y - m(X))(m(X) - \mu)] && \text{(expand the square; linearity of expectation)} \end{aligned}$$

The three terms are handled separately, each using the **law of iterated expectations**, applied to a function of X and Y (**conditional expectation of a function**). The third

term also uses two facts about conditional expectations: a [function of \$X\$ factors out of a conditional expectation given \$X\$](#) , and [conditional expectation is linear](#).

First term:

$$\begin{aligned} E[(Y - m(X))^2] &= E[E[(Y - m(X))^2 \mid X]] && \text{(law of iterated expectations)} \\ &= E[\text{Var}(Y \mid X)] && \text{(definition of conditional variance)} \end{aligned}$$

Second term: by the law of iterated expectations, $E[m(X)] = E[E[Y \mid X]] = \mu$, so:

$$\begin{aligned} E[(m(X) - \mu)^2] &= E[(m(X) - E[m(X)])^2] && (\mu = E[m(X)]) \\ &= \text{Var}(m(X)) && \text{(definition of variance)} \\ &= \text{Var}(E[Y \mid X]) && \text{(definition of } m(X)) \end{aligned}$$

Third term:

$$\begin{aligned} E[(Y - m(X))[m(X) - \mu]] &= E[E[(Y - m(X))[m(X) - \mu] \mid X]] && \text{(law of iterated expectations)} \\ &= E[(m(X) - \mu) \cdot E[Y - m(X) \mid X]] && \text{(a function of } X \text{ factors out, with } g(X) = E[Y - m(X) \mid X]) \\ &= E[(m(X) - \mu) \cdot (E[Y \mid X] - E[m(X) \mid X])] && \text{(linearity of conditional expectation)} \\ &= E[(m(X) - \mu) \cdot (E[Y \mid X] - m(X))] && (E[g(X) \mid X] = g(X), \text{ with } g = m) \\ &= E[(m(X) - \mu) \cdot 0] && (E[Y \mid X] = m(X)) \\ &= 0 && \text{(expectation of a constant)} \end{aligned}$$

Substituting the three terms:

$$\begin{aligned} \text{Var}(Y) &= E[\text{Var}(Y \mid X)] + \text{Var}(E[Y \mid X]) + 2 \cdot 0 && \text{(substitute the three terms)} \\ &= E[\text{Var}(Y \mid X)] + \text{Var}(E[Y \mid X]) && \text{(simplify)} \end{aligned}$$

□

Alternate names include: the **conditional variance formula**, **Eve's law**, and the **variance decomposition formula**.

Example 0.7 (Decomposing the variance of a binary outcome). Continuing Example 0.4, $P(X = 0) = P(X = 1) = 0.5$, $E[Y \mid X = 0] = 0.6$, and $E[Y \mid X = 1] = 0.8$, so $E[E[Y \mid X]] = 0.7$, and by Theorem 0.3:

$$\begin{aligned} E[\text{Var}(Y \mid X)] &= 0.5 \cdot 0.24 + 0.5 \cdot 0.16 = 0.20 && \text{(average the conditional variances)} \\ \text{Var}(E[Y \mid X]) &= 0.5 \cdot (0.6 - 0.7)^2 + 0.5 \cdot (0.8 - 0.7)^2 = 0.01 && \text{(variance of the conditional means)} \\ \text{Var}(Y) &= 0.20 + 0.01 = 0.21 && \text{(law of total variance)} \end{aligned}$$

As a check, Y is Bernoulli with $P(Y = 1) = 0.3 + 0.4 = 0.7$, so by Example 0.3, $\text{Var}(Y) = 0.7 \cdot 0.3 = 0.21$.

Definition 0.7 (Precision). The **precision** of a random variable X , often denoted $\tau(X)$, τ_X , or shorthand as τ , is the inverse of that random variable's **variance** (for $\text{Var}(X) > 0$); that is:

$$\tau(X) \stackrel{\text{def}}{=} (\text{Var}(X))^{-1}$$

Definition 0.8 (Standard deviation). The **standard deviation** of a random variable X is the square root of the **variance** of X :

$$\text{SD}(X) \stackrel{\text{def}}{=} \sqrt{\text{Var}(X)}$$

The standard deviation is on the same scale as X itself (unlike the variance, which is on the scale of X 's square), which is why it is often the preferred measure of spread when reporting results.

Example 0.8 (Precision and standard deviation of a fair coin flip). In Example 0.3, a fair coin flip has $\text{Var}(X) = 1/4$, so its precision is $\tau(X) = 1/(1/4) = 4$ and its standard deviation is $\text{SD}(X) = \sqrt{1/4} = 1/2$.

Covariance

Definition 0.9 (Covariance). The **covariance** of two random variables X and Y with $E[X^2] < \infty$ and $E[Y^2] < \infty$ is:

$$\text{Cov}(X, Y) \stackrel{\text{def}}{=} E[(X - E[X])(Y - E[Y])]$$

Theorem 0.4 (Alternative formula for covariance).

$$\text{Cov}(X, Y) = E[XY] - E[X]E[Y]$$

Proof

Proof. By **linearity of expectation**, analogous to the proof of the **simplified expression for variance**:

$$\begin{aligned}
\text{Cov}(X, Y) &\stackrel{\text{def}}{=} E[(X - E[X])(Y - E[Y])] && \text{(definition of covariance)} \\
&= E[XY - X E[Y] - Y E[X] + E[X] E[Y]] && \text{(expand the product)} \\
&= E[XY] - E[X] E[Y] - E[Y] E[X] + E[X] E[Y] && \text{(linearity of expectation; } E[X], E[Y] \text{ are constants)} \\
&= E[XY] - E[X] E[Y] && \text{(combine like terms)}
\end{aligned}$$

□

Example 0.9 (Covariance of a binary exposure and outcome). For the joint PMF in Example 0.4, $E[XY] = P(X = 1, Y = 1) = 0.4$, $E[X] = 0.5$, and $E[Y] = 0.7$, so:

$$\begin{aligned}
\text{Cov}(X, Y) &= E[XY] - E[X] E[Y] && \text{(alternative formula for covariance)} \\
&= 0.4 - 0.5 \cdot 0.7 && \text{(substitute)} \\
&= 0.05 && \text{(evaluate)}
\end{aligned}$$

Theorem 0.5 (Independent random variables have zero covariance). *If X and Y are independent, each discrete or continuous, with defined expectations, then:*

$$E[XY] = E[X] E[Y]$$

If also $E[X^2] < \infty$ and $E[Y^2] < \infty$, then $\text{Cov}(X, Y) = 0$.

Proof

Proof. Write f_X and f_Y for the PMFs or densities of X and Y , with reference measures μ_X and μ_Y as in the [joint-distribution form of Fubini–Tonelli](#) (counting measure for a discrete variable, Lebesgue measure for a continuous one). Because X and Y are independent, $f_{X,Y}(x, y) = f_X(x) f_Y(y)$ is their joint PMF, density, or density-mass function (the factorization in the notes to the [definition of independence](#)). First, with $h(x, y) = |x||y| \geq 0$ (condition (a)):

$$\begin{aligned}
E[|XY|] &= \int \left(\int |x||y| f_X(x) f_Y(y) d\mu_Y(y) \right) d\mu_X(x) && \text{(joint-distribution form of Fubini–Tonelli, condition (a))} \\
&= \int |x| f_X(x) \left(\int |y| f_Y(y) d\mu_Y(y) \right) d\mu_X(x) && (|x| f_X(x) \text{ does not depend on } y) \\
&= \int |x| f_X(x) \cdot E[|Y|] d\mu_X(x) && \text{(LOTUS for } |Y|) \\
&= E[|X|] \cdot E[|Y|] && \text{(LOTUS for } |X|)
\end{aligned}$$

which is finite, because X and Y have defined expectations. So condition (b) holds for

$h(x, y) = xy$, and the same steps without the absolute values give:

$$\begin{aligned}
E[XY] &= \int \left(\int xy f_X(x) f_Y(y) d\mu_Y(y) \right) d\mu_X(x) && \text{(joint-distribution form of Fubini–Tonelli, condition (b))} \\
&= \int x f_X(x) \left(\int y f_Y(y) d\mu_Y(y) \right) d\mu_X(x) && (x f_X(x) \text{ does not depend on } y) \\
&= \int x f_X(x) \cdot E[Y] d\mu_X(x) && \text{(definition of expectation)} \\
&= E[X] \cdot E[Y] && \text{(definition of expectation)}
\end{aligned}$$

Finally, by the [alternative formula for covariance](#):

$$\begin{aligned}
\text{Cov}(X, Y) &= E[XY] - E[X] E[Y] && \text{(alternative formula for covariance)} \\
&= E[X] E[Y] - E[X] E[Y] && (E[XY] = E[X] E[Y]) \\
&= 0 && \text{(subtract)}
\end{aligned}$$

□

Example 0.10 (Two independent coin flips). For the independent coin flips X_1 and X_2 of [the independence page's example](#), $X_1 X_2 = 1$ only when both flips are heads, so:

$$\begin{aligned}
E[X_1 X_2] &= P(X_1 = 1, X_2 = 1) && (X_1 X_2 \text{ is the indicator of two heads}) \\
&= \frac{1}{4} && \text{(each outcome has probability } \frac{1}{4} \text{)} \\
&= \frac{1}{2} \cdot \frac{1}{2} && \text{(factor)} \\
&= E[X_1] E[X_2] && \text{(each flip is Ber}(1/2)\text{)}
\end{aligned}$$

so $\text{Cov}(X_1, X_2) = 0$, as Theorem [0.5](#) requires.

Example 0.11 (Zero covariance without independence). The converse of Theorem [0.5](#) is false. Let X take the values $-1, 0$, and 1 with probability $1/3$ each, and let $Y = X^2$. Then $E[X] = (-1 + 0 + 1)/3 = 0$, and $XY = X^3 = X$, so:

$$\begin{aligned}
\text{Cov}(X, Y) &= E[XY] - E[X] E[Y] && \text{(alternative formula for covariance)} \\
&= E[X] - E[X] E[Y] && (XY = X^3 = X \text{ on } \{-1, 0, 1\}) \\
&= 0 - 0 \cdot E[Y] && (E[X] = 0) \\
&= 0 && \text{(multiply)}
\end{aligned}$$

But X and Y are not independent: Y is a function of X , and $P(X = 0, Y = 0) = P(X = 0) = \frac{1}{3}$, while $P(X = 0) P(Y = 0) = \frac{1}{3} \cdot \frac{1}{3} = \frac{1}{9}$.

Definition 0.10 (Correlation). The **correlation** of two random variables X and Y with finite, positive **variances** is their **covariance** divided by the product of their **standard deviations**:

$$\text{Cor}(X, Y) \stackrel{\text{def}}{=} \frac{\text{Cov}(X, Y)}{\text{SD}(X) \text{SD}(Y)}$$

Dividing by the standard deviations removes the units of X and Y , and the correlation always lies in $[-1, 1]$ (Casella and Berger 2002). Random variables with zero covariance, and hence zero correlation, are called **uncorrelated**; Theorem 0.5 says independent random variables are uncorrelated, and Example 0.11 shows the converse fails. This population correlation is a property of a joint distribution; the sample (Pearson) correlation coefficient computed from data estimates it.

Example 0.12 (Correlation of a binary exposure and outcome). In Example 0.9, $\text{Cov}(X, Y) = 0.05$, where $X \sim \text{Ber}(0.5)$ has $\text{Var}(X) = 0.25$ (Example 0.3) and Y has $\text{Var}(Y) = 0.21$ (Example 0.7), so:

$$\begin{aligned} \text{Cor}(X, Y) &= \frac{\text{Cov}(X, Y)}{\text{SD}(X) \text{SD}(Y)} && \text{(definition of correlation)} \\ &= \frac{0.05}{\sqrt{0.25} \cdot \sqrt{0.21}} && \text{(substitute; } \text{SD}(\cdot) = \sqrt{\text{Var}(\cdot)} \text{)} \\ &\approx \frac{0.05}{0.5 \cdot 0.458} && \text{(evaluate the square roots)} \\ &\approx 0.218 && \text{(divide)} \end{aligned}$$

Definition 0.11 (Conditional covariance). The **conditional covariance** of Y and Z given $X = x$ is their covariance under their conditional distribution given $X = x$:

$$\text{Cov}(Y, Z \mid X = x) \stackrel{\text{def}}{=} \text{E}[(Y - \text{E}[Y \mid X = x])(Z - \text{E}[Z \mid X = x]) \mid X = x]$$

Evaluating this function of x at X gives the random variable $\text{Cov}(Y, Z \mid X)$.

Example 0.13 (Conditional covariance of a variable with itself). Taking $Z = Y$ in Definition 0.11 gives the **conditional variance**: $\text{Cov}(Y, Y \mid X = x) = \text{Var}(Y \mid X = x)$. In Example 0.4, for example, $\text{Cov}(Y, Y \mid X = 0) = 0.24$.

Theorem 0.6 (Law of total covariance). *For random variables X , Y , and Z with $\text{E}[Y^2] < \infty$ and $\text{E}[Z^2] < \infty$:*

$$\text{Cov}(Y, Z) = \text{E}[\text{Cov}(Y, Z \mid X)] + \text{Cov}(\text{E}[Y \mid X], \text{E}[Z \mid X])$$

i Proof

Proof. The proof follows the proof of the [law of total variance](#), which is the special case $Z = Y$. Write $m_Y(X) \stackrel{\text{def}}{=} \text{E}[Y \mid X]$, $m_Z(X) \stackrel{\text{def}}{=} \text{E}[Z \mid X]$, $\mu_Y \stackrel{\text{def}}{=} \text{E}[Y]$, and $\mu_Z \stackrel{\text{def}}{=} \text{E}[Z]$. Adding and subtracting $m_Y(X)$ and $m_Z(X)$ inside the deviations:

$$\begin{aligned} \text{Cov}(Y, Z) &= \text{E}[(Y - \mu_Y)(Z - \mu_Z)] && \text{(definition of covariance)} \\ &= \text{E}[(Y - m_Y(X) + m_Y(X) - \mu_Y)(Z - m_Z(X) + m_Z(X) - \mu_Z)] && \text{(add and subtract)} \\ &= \text{E}[(Y - m_Y(X))(Z - m_Z(X))] + \text{E}[(Y - m_Y(X))(m_Z(X) - \mu_Z)] \\ &\quad + \text{E}[(m_Y(X) - \mu_Y)(Z - m_Z(X))] + \text{E}[(m_Y(X) - \mu_Y)(m_Z(X) - \mu_Z)] && \text{(expand the product; linearity)} \end{aligned}$$

The two middle terms are 0, by the same steps as the cross term in the proof of Theorem 0.3: condition on X by the [law of iterated expectations](#), factor out the function of X ([pull-out property](#)), and use $\text{E}[Y - m_Y(X) \mid X] = 0$ (or $\text{E}[Z - m_Z(X) \mid X] = 0$), which follows from [linearity of conditional expectation](#). For the first term:

$$\begin{aligned} \text{E}[(Y - m_Y(X))(Z - m_Z(X))] &= \text{E}[\text{E}[(Y - m_Y(X))(Z - m_Z(X)) \mid X]] && \text{(law of iterated expectations)} \\ &= \text{E}[\text{Cov}(Y, Z \mid X)] && \text{(definition of conditional covariance)} \end{aligned}$$

For the last term, the law of iterated expectations gives $\text{E}[m_Y(X)] = \mu_Y$ and $\text{E}[m_Z(X)] = \mu_Z$, so:

$$\begin{aligned} \text{E}[(m_Y(X) - \mu_Y)(m_Z(X) - \mu_Z)] &= \text{E}[(m_Y(X) - \text{E}[m_Y(X)])(m_Z(X) - \text{E}[m_Z(X)])] && \text{(substitute the means)} \\ &= \text{Cov}(m_Y(X), m_Z(X)) && \text{(definition of covariance)} \\ &= \text{Cov}(\text{E}[Y \mid X], \text{E}[Z \mid X]) && \text{(definitions of } m_Y, m_Z) \end{aligned}$$

Adding the four terms gives the result. $\square$

Alternate names include: the **covariance decomposition formula** and the **conditional covariance formula**.

Lemma 0.1 (The covariance of a variable with itself is its variance). *For any random variable X with $\text{E}[X^2] < \infty$:*

$$\text{Cov}(X, X) = \text{Var}(X)$$

i Proof

Proof. By the [alternative formula for covariance](#):

$$\begin{aligned}\text{Cov}(X, X) &= \text{E}[XX] - \text{E}[X] \text{E}[X] && \text{(alternative formula for covariance, with } Y = X) \\ &= \text{E}[X^2] - (\text{E}[X])^2 && (XX = X^2) \\ &= \text{Var}(X) && \text{(simplified expression for variance)}\end{aligned}$$

□

Definition 0.12 (Variance/covariance of a $p \times 1$ random vector). For a $p \times 1$ dimensional random vector $\tilde{X}$,

$$\begin{aligned}\text{Var}(\tilde{X}) &\stackrel{\text{def}}{=} \text{Cov}(\tilde{X}) \\ &\stackrel{\text{def}}{=} \text{E}\left[(\tilde{X} - \text{E} \tilde{X})(\tilde{X} - \text{E} \tilde{X})^\top\right]\end{aligned}$$

Theorem 0.7 (Elements of the variance-covariance matrix are pairwise covariances). For a $p \times 1$ random vector $\tilde{X} = (X_1, \dots, X_p)^\top$, the (i, j) -th element of $\text{Var}(\tilde{X})$ is $\text{Cov}(X_i, X_j)$:

$$\text{Var}(\tilde{X}) = \begin{pmatrix} \text{Var}(X_1) & \text{Cov}(X_1, X_2) & \cdots & \text{Cov}(X_1, X_p) \\ \text{Cov}(X_2, X_1) & \text{Var}(X_2) & \cdots & \text{Cov}(X_2, X_p) \\ \vdots & \vdots & \ddots & \vdots \\ \text{Cov}(X_p, X_1) & \text{Cov}(X_p, X_2) & \cdots & \text{Var}(X_p) \end{pmatrix}$$

i Proof

Proof. Let $\mu_i = \text{E}[X_i]$ for $i = 1, \dots, p$, so $\text{E} \tilde{X} = (\mu_1, \dots, \mu_p)^\top$. By Definition [0.12](#):

$$\begin{aligned}
\text{Var}(\tilde{X}) &= \mathbb{E}[(\tilde{X} - \mathbb{E} \tilde{X})(\tilde{X} - \mathbb{E} \tilde{X})^\top] \\
&= \mathbb{E} \left[\begin{pmatrix} X_1 - \mu_1 \\ \vdots \\ X_p - \mu_p \end{pmatrix} (X_1 - \mu_1 \quad \cdots \quad X_p - \mu_p) \right] \\
&= \mathbb{E} \left[\begin{pmatrix} (X_1 - \mu_1)(X_1 - \mu_1) & \cdots & (X_1 - \mu_1)(X_p - \mu_p) \\ \vdots & \ddots & \vdots \\ (X_p - \mu_p)(X_1 - \mu_1) & \cdots & (X_p - \mu_p)(X_p - \mu_p) \end{pmatrix} \right] \\
&= \begin{pmatrix} \mathbb{E}[(X_1 - \mu_1)(X_1 - \mu_1)] & \cdots & \mathbb{E}[(X_1 - \mu_1)(X_p - \mu_p)] \\ \vdots & \ddots & \vdots \\ \mathbb{E}[(X_p - \mu_p)(X_1 - \mu_1)] & \cdots & \mathbb{E}[(X_p - \mu_p)(X_p - \mu_p)] \end{pmatrix} \\
&= \begin{pmatrix} \text{Cov}(X_1, X_1) & \cdots & \text{Cov}(X_1, X_p) \\ \vdots & \ddots & \vdots \\ \text{Cov}(X_p, X_1) & \cdots & \text{Cov}(X_p, X_p) \end{pmatrix} \\
&= \begin{pmatrix} \text{Var}(X_1) & \cdots & \text{Cov}(X_1, X_p) \\ \vdots & \ddots & \vdots \\ \text{Cov}(X_p, X_1) & \cdots & \text{Var}(X_p) \end{pmatrix}
\end{aligned}$$

where:

- the step from the third to fourth line uses the [expectation of a random matrix](#),
- the step from the fourth to fifth line uses Definition 0.9, and
- the last step uses Lemma 0.1.

□

Theorem 0.8 (Alternate expression for variance of a random vector).

$$\text{Var}(\tilde{X}) = \mathbb{E}[\tilde{X}\tilde{X}^\top] - (\mathbb{E} \tilde{X})(\mathbb{E} \tilde{X})^\top$$

i Proof

Proof.

$$\begin{aligned}
\text{Var}(\tilde{X}) &= \mathbb{E}[(\tilde{X} - \mathbb{E} \tilde{X})(\tilde{X} - \mathbb{E} \tilde{X})^\top] && \text{(definition)} \\
&= \mathbb{E}[\tilde{X}\tilde{X}^\top - \tilde{X}(\mathbb{E} \tilde{X})^\top - (\mathbb{E} \tilde{X})\tilde{X}^\top + (\mathbb{E} \tilde{X})(\mathbb{E} \tilde{X})^\top] && \text{(expand the product)} \\
&= \mathbb{E}[\tilde{X}\tilde{X}^\top] - (\mathbb{E} \tilde{X})(\mathbb{E} \tilde{X})^\top - (\mathbb{E} \tilde{X})(\mathbb{E} \tilde{X})^\top + (\mathbb{E} \tilde{X})(\mathbb{E} \tilde{X})^\top && \text{(linearity, element-wise; } \mathbb{E} \tilde{X} \text{ is constant)} \\
&= \mathbb{E}[\tilde{X}\tilde{X}^\top] - (\mathbb{E} \tilde{X})(\mathbb{E} \tilde{X})^\top && \text{(combine like terms)}
\end{aligned}$$

□

Theorem 0.9 (Variance of a linear combination). *For any vector of random variables $\tilde{X} = (X_1, \dots, X_n)$ and corresponding vector of constants $\tilde{a} = (a_1, \dots, a_n)$, the variance of their linear combination is:*

$$\begin{aligned}\text{Var}(\tilde{a} \cdot \tilde{X}) &= \text{Var}\left(\sum_{i=1}^n a_i X_i\right) \\ &= \tilde{a}^\top \text{Var}(\tilde{X}) \tilde{a} \\ &= \sum_{i=1}^n \sum_{j=1}^n a_i a_j \text{Cov}(X_i, X_j)\end{aligned}$$

i Proof

Proof. Treat $\tilde{a}$ and $\tilde{X}$ as $n \times 1$ column vectors, so $\tilde{a} \cdot \tilde{X} = \tilde{a}^\top \tilde{X} = \sum_{i=1}^n a_i X_i$, a scalar. By linearity of expectation, $\mathbb{E}[\tilde{a}^\top \tilde{X}] = \tilde{a}^\top \mathbb{E} \tilde{X}$, so:

$$\begin{aligned}\text{Var}(\tilde{a}^\top \tilde{X}) &= \mathbb{E}\left[(\tilde{a}^\top \tilde{X} - \tilde{a}^\top \mathbb{E} \tilde{X})^2\right] && \text{(definition of variance)} \\ &= \mathbb{E}\left[(\tilde{a}^\top (\tilde{X} - \mathbb{E} \tilde{X}))^2\right] && \text{(factor out } \tilde{a}^\top) \\ &= \mathbb{E}\left[\tilde{a}^\top (\tilde{X} - \mathbb{E} \tilde{X}) (\tilde{X} - \mathbb{E} \tilde{X})^\top \tilde{a}\right] && \text{(a scalar equals its transpose, so } s^2 = s s^\top) \\ &= \tilde{a}^\top \mathbb{E}\left[(\tilde{X} - \mathbb{E} \tilde{X}) (\tilde{X} - \mathbb{E} \tilde{X})^\top\right] \tilde{a} && \text{(linearity of expectation, element-wise)} \\ &= \tilde{a}^\top \text{Var}(\tilde{X}) \tilde{a} && \text{(variance of a random vector)} \\ &= \sum_{i=1}^n \sum_{j=1}^n a_i a_j \text{Cov}(X_i, X_j) && \text{(expand the quadratic form, using the elements of } \text{Var}(\tilde{X}))\end{aligned}$$

□

Corollary 0.1 (Variance of a sum of two random variables). *For any two random variables X and Y and scalars a and b :*

$$\text{Var}(aX + bY) = a^2 \text{Var}(X) + b^2 \text{Var}(Y) + 2(a \cdot b) \text{Cov}(X, Y)$$

Proof

Proof. Apply Theorem 0.9 with $n = 2$, $X_1 = X$, and $X_2 = Y$:

$$\text{Var}(aX + bY) = a^2 \text{Var}(X) + b^2 \text{Var}(Y) + 2ab \text{Cov}(X, Y)$$

Alternatively, by linearity of expectation:

$$\begin{aligned} \text{Var}(aX + bY) &\stackrel{\text{def}}{=} \text{E}[(aX + bY - \text{E}[aX + bY])^2] && \text{(definition of variance)} \\ &= \text{E}[(a(X - \text{E}[X]) + b(Y - \text{E}[Y]))^2] && \text{(linearity of expectation)} \\ &= \text{E}[a^2(X - \text{E}[X])^2 + 2(a \cdot b)(X - \text{E}[X])(Y - \text{E}[Y]) + b^2(Y - \text{E}[Y])^2] && \text{(expand the square)} \\ &= a^2 \text{E}[(X - \text{E}[X])^2] + 2(a \cdot b) \text{E}[(X - \text{E}[X])(Y - \text{E}[Y])] + b^2 \text{E}[(Y - \text{E}[Y])^2] && \text{(linearity of expectation)} \\ &= a^2 \text{Var}(X) + 2(a \cdot b) \text{Cov}(X, Y) + b^2 \text{Var}(Y) && \text{(definitions of variance and covariance)} \end{aligned}$$

□

This corollary is why two variables' covariance matters for combining them: if X and Y are **independent**, then $\text{Cov}(X, Y) = 0$ (Theorem 0.5), and the variance of their sum is just the sum of their variances.

References

Casella, George, and Roger Berger. 2002. *Statistical Inference*. 2nd ed. Cengage Learning. <https://www.cengage.com/c/statistical-inference-2e-casella-berger/9780534243128/>.